From Nothingness, Everything: Finite Causal Diamonds, the Emergence of Quantum Mechanics, and Mass Generation at the Schwarzschild Horizon

March 2026

Abstract

We study the causal structure of the Schwarzschild black hole interior and its implications for the domain of applicability of quantum mechanics. We prove that the maximal causal diamond that can be embedded in the future of any event inside the horizon has finite four-volume, bounded by the proper time to the singularity, and that this volume shrinks monotonically to zero at $r = 0$. This is a coordinate-invariant, physical distinction from the exterior, where the causal future is infinite and causal diamonds of arbitrary size exist.

We show that while the Einstein equivalence principle guarantees the local validity of quantum field theory inside the horizon, there is a *global* constraint with no exterior analogue: every physical process in the interior has a finite remaining lifetime. Objects whose internal dynamics require more time than is available cannot be sustained. This defines a hierarchy of causal viability that is trivially satisfied in the exterior (where unlimited time is available) but becomes progressively restrictive toward the singularity.

We introduce a covariantly defined scalar $\mathcal{X} = g^{\mu\nu}\nabla_\mu\mathcal{R}\,\nabla_\nu\mathcal{R}$, the norm of the gradient of the areal radius $\mathcal{R}$, which is positive in the exterior, zero at the horizon, and negative in the interior. We construct a $U(1)_\mathcal{C}$ gauge theory with an effective potential $V(|\varphi|;\mathcal{X})$ that drives spontaneous symmetry breaking when $\mathcal{X} > 0$ and restores the symmetry when $\mathcal{X} \leq 0$. The resulting position-dependent vacuum expectation value and gauge boson mass are *derived* from the equations of motion, not postulated. We propose that this causal Higgs mechanism governs the emergence of mass and quantum structure at the horizon boundary.

Neither general relativity nor quantum mechanics is modified or contradicted. What we identify is their hierarchical relationship: gravity determines the causal structure from which quantum mechanics emerges. This is work is first of a series developing these ideas.

Keywords: Schwarzschild spacetime, causal diamonds, causal structure, Higgs mechanism, mass generation, information

Contents

1 Introduction **4**
- 1.1 Motivation . . . 4
- 1.2 The key distinction: finite versus infinite causal futures . . . 4
- 1.3 What is and is not claimed . . . 4

2 The Schwarzschild Geometry **5**

3 Causal Diamonds and the Finite Interior **5**
- 3.1 Why equal-time slices are not the point . . . 5
- 3.2 What is special: the finite causal future . . . 5
- 3.3 Comparison with Minkowski space . . . 7

4 The Equivalence Principle and Local Quantum Field Theory **7**
- 4.1 Local physics is standard . . . 7
- 4.2 The global constraint: finite lifetime of all interior processes . . . 7
- 4.3 Objects do not dissolve instantaneously; they become unsustainable . . . 8

5 The Hierarchy: Gravity and Quantum Mechanics **8**
- 5.1 What survives at the singularity . . . 8
- 5.2 The hierarchy . . . 9
- 5.3 On the quantizability of gravity . . . 9

6 Nothingness as the Limit of Vanishing Causal Diamonds **9**

7 The Covariant Causal Gauge Theory **10**
- 7.1 The trapped/untrapped scalar . . . 10
- 7.2 Field content . . . 11
- 7.3 The geometry-coupled Lagrangian . . . 11

8 The Causal Higgs Mechanism: Stepwise Derivation **11**
- 8.1 Step 1: The effective mass and the vacuum structure . . . 11
- 8.2 Step 2: The vacuum expectation value (derived) . . . 12
- 8.3 Step 3: Consistency with the equation of motion . . . 12
- 8.4 Step 4: The gauge boson mass . . . 13
- 8.5 Step 5: The Higgs (radial) mode mass . . . 13
- 8.6 Step 6: Summary of the mechanism . . . 13

9 Mass Creation at the Horizon: Phenomenological Framework **14**
- 9.1 The gradient energy of the condensate . . . 14
- 9.2 The static solution paradox and the need for dynamics . . . 14
- 9.3 The phenomenological ansatz . . . 14
- 9.4 Consequences of the ansatz . . . 15

10 On Hawking Radiation in This Framework **15**
- 10.1 The standard derivation . . . 15
- 10.2 Framework-dependent assumptions . . . 15

11 Information **15**
11.1 Information and the causal condensate 15
11.2 The information paradox 16

12 The Causality Destruction Theorem **16**
12.1 Setup 16
12.2 The theorem 16

13 Summary **17**

14 Discussion **17**
14.1 What this framework does and does not claim 17
14.2 What the covariant formulation achieves 18
14.3 What the mass creation ansatz needs 18
14.4 Testable predictions 18
14.5 Open problems 18

15 Conclusion **19**

1 Introduction

1.1 Motivation

A central problem of fundamental physics has been the reconciliation of general relativity and quantum mechanics [4, 5]. The standard approach assumes that gravity must ultimately be described by a quantum theory. String theory [6], loop quantum gravity [7], and asymptotic safety [8] all proceed from this assumption.

We propose a re-examination of this assumption, inspired by Dyson's observation [9] that the correct direction may be gravito-quantum—understanding how quantum phenomena emerge from the gravitational arena—rather than quantum gravity. Both general relativity and quantum mechanics are experimentally confirmed theories. Neither is refuted by this work. What we offer is a clarification of their *relationship*: general relativity determines the causal structure of spacetime, and quantum mechanics requires that causal structure as a precondition for its operation.

1.2 The key distinction: finite versus infinite causal futures

In *any* spacetime, including flat Minkowski space, a Cauchy surface (equal-time hypersurface) is spacelike, and no causal curve connects distinct events on it. This is not a pathology; it is how spacetime is sliced. Quantum field theory is built on spacelike Cauchy surfaces: equal-time commutation relations enforce spacelike separation, and coherence is maintained by Hamiltonian evolution *between* successive slices, not within a single slice.

The Schwarzschild interior is *not* unusual because its equal-time slices are spacelike—all equal-time slices everywhere are spacelike. It is unusual because:

> *The causal future of every event inside the horizon is finite. It terminates at the singularity in bounded proper time. In the exterior, the causal future is infinite.*

This distinction—finite versus infinite causal future—is coordinate-invariant, physically meaningful, and has no analogue in Minkowski space or in the exterior Schwarzschild geometry. It is the foundation of everything that follows.

1.3 What is and is not claimed

We make several claims of different logical strength:

Rigorous results (follow from the Schwarzschild metric alone):

- The maximal causal diamond in the interior has finite four-volume, bounded by $\tau_{\text{coh}}(r)$, and shrinks to zero at the singularity (section 3).
- The total exterior causal content decreases monotonically when the horizon area increases (section 12).

Physical arguments (follow from the rigorous results plus standard physics):

- The equivalence principle guarantees local QFT validity inside the horizon, but global features (stable bound states, indefinite coherence) require infinite causal future and are therefore exterior-only phenomena (section 4).

- This establishes a hierarchy: gravity → causal structure → quantum mechanics (section 5).

Theoretical framework (postulated, with derivations from the postulates):

- A covariantly coupled causal gauge theory in which the VEV is *derived* from the equations of motion (sections 7 and 8).
- A phenomenological mass creation ansatz, whose proper derivation requires a dynamical spacetime (section 9).

2 The Schwarzschild Geometry

The Schwarzschild line element in coordinates (t, r, θ, φ) is

$$ds^2 = -f(r)\, c^2\, dt^2 + f(r)^{-1}\, dr^2 + r^2\, d\Omega^2\,, \tag{1}$$

where $f(r) = 1 - r_{\rm s}/r$, $r_{\rm s} = 2GM/c^2$, and $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$.

Exterior ($r > r_{\rm s}$): $f > 0$. The coordinate t is timelike.

Interior ($0 < r < r_{\rm s}$): $f < 0$. The coordinate r is timelike; the singularity at $r = 0$ lies in every observer's future.

Horizon ($r = r_{\rm s}$): $f = 0$. The surface is null. The coordinate singularity is removable [2]; the geometry is smooth.

For all subsequent analysis, the key object is the *areal radius* $\mathcal{R}$, defined geometrically: on any two-sphere of symmetry, $\mathcal{R}$ is the value such that the area equals $4\pi\mathcal{R}^2$. In Schwarzschild coordinates, $\mathcal{R} = r$, but $\mathcal{R}$ is a *coordinate scalar*—a geometric quantity defined invariantly.

3 Causal Diamonds and the Finite Interior

3.1 Why equal-time slices are not the point

In any Lorentzian spacetime, a spacelike hypersurface has a positive-definite induced metric. No causal curve connects distinct events on such a surface. This is true in Minkowski space, in the Schwarzschild exterior, and in the Schwarzschild interior. It is a general property of spatial slices, not a special feature of the interior.

The dynamical content of any physical theory resides in evolution *between* successive slices (Cauchy evolution), not in connections *within* a single slice. In the interior, the relevant time coordinate is r (which decreases toward the future), and the Cauchy problem from one constant-r slice to the next is locally well-posed [16].

Therefore, the causal disconnectedness of individual constant-r slices does not, by itself, imply any failure of physical coherence or quantum mechanics.

3.2 What is special: the finite causal future

The genuine and coordinate-invariant distinction between the interior and the exterior is:

Proposition 3.1 (Finite causal diamonds in the interior)**.** *Let p be an event in the Schwarzschild interior at areal radius $\mathcal{R}_0 < r_\mathrm{s}$. The causal future $J^+(p)$ of p is compact (it terminates at the singularity $\mathcal{R} = 0$). The maximal proper time from p to the singularity, along any future-directed causal curve, is bounded by*

$$\tau_{\max}(\mathcal{R}_0) \leq \frac{\pi r_\mathrm{s}}{2c}\,. \tag{2}$$

The maximal causal diamond[1] *that fits within the remaining future of p has finite four-volume $\mathrm{Vol}_4(D)$ satisfying*

$$\mathrm{Vol}_4(D) \;\longrightarrow\; 0 \quad \textit{as} \quad \mathcal{R}_0 \to 0\,. \tag{3}$$

Proof. That the future terminates follows from the Penrose singularity theorem [3]: the interior contains a trapped surface, and all future-directed causal curves from p reach the singularity in finite affine parameter. For a radial free-fall geodesic from rest at $\mathcal{R}_0$, the proper time to $\mathcal{R} = 0$ is (see derivation below)

$$\tau_{\mathrm{coh}}(\mathcal{R}_0) = \frac{r_\mathrm{s}}{c}\left[\arccos\left(\sqrt{\frac{\mathcal{R}_0}{r_\mathrm{s}}}\right) + \sqrt{\frac{\mathcal{R}_0}{r_\mathrm{s}}\left(1 - \frac{\mathcal{R}_0}{r_\mathrm{s}}\right)}\,\right], \tag{4}$$

which satisfies $\tau_{\mathrm{coh}}(r_\mathrm{s}) = \pi r_\mathrm{s}/(2c)$ and $\tau_{\mathrm{coh}}(0) = 0$.

The spatial extent of the causal diamond is bounded by $c\,\tau_{\mathrm{coh}}$ (no signal can travel farther than $c\,\tau_{\mathrm{coh}}$ before the singularity). As $\mathcal{R}_0 \to 0$, $\tau_{\mathrm{coh}} \to 0$, and the causal diamond collapses to zero volume. □

Derivation of (4). For a radial free-fall geodesic in the Schwarzschild geometry, the equation of motion gives [16]

$$\left(\frac{d\mathcal{R}}{d\tau}\right)^2 = \frac{r_\mathrm{s}\,c^2}{\mathcal{R}}\,. \tag{5}$$

Since $\mathcal{R}$ decreases, $d\mathcal{R}/d\tau = -c\sqrt{r_\mathrm{s}/\mathcal{R}}$. Separating variables:

$$d\tau = -\frac{1}{c}\sqrt{\frac{\mathcal{R}}{r_\mathrm{s}}}\,d\mathcal{R}\,. \tag{6}$$

Integrating from $\mathcal{R}_0$ to 0, substituting $\mathcal{R} = r_\mathrm{s}\cos^2\eta$:

$$\tau_{\mathrm{coh}} = \frac{1}{c}\int_0^{\mathcal{R}_0}\sqrt{\frac{\mathcal{R}}{r_\mathrm{s}}}\,d\mathcal{R} = \frac{r_\mathrm{s}}{c}\int_0^{\eta_0} 2\cos^2\eta\,d\eta = \frac{r_\mathrm{s}}{c}\left[\eta_0 + \frac{1}{2}\sin 2\eta_0\right], \tag{7}$$

where $\cos\eta_0 = \sqrt{\mathcal{R}_0/r_\mathrm{s}}$, yielding (4). □

Remark 3.2. *In the exterior ($\mathcal{R} > r_\mathrm{s}$), the causal future of any event is infinite: future-directed timelike curves can be extended to infinite proper time. Causal diamonds of arbitrary size and duration can be constructed. This is the fundamental asymmetry between interior and exterior. In Minkowski space, the causal future is likewise infinite at every event.*

[1] The causal diamond $D(p, q)$ of two timelike-separated events p and q is the intersection $J^+(p) \cap J^-(q)$.

3.3 Comparison with Minkowski space

The reviewer's objection that "the same logic would dissolve objects in Minkowski space" is addressed precisely by this section. In Minkowski space, at every event:

- The causal future is infinite.
- Causal diamonds of arbitrary size and duration exist.
- There is no upper bound on the time available for physical processes.

In the Schwarzschild interior, at every event:

- The causal future is finite (terminates at the singularity).
- Causal diamonds are bounded by $\tau_{\text{coh}}(\mathcal{R})$.
- As $\mathcal{R} \to 0$, the bound shrinks to zero.

The distinction is not about individual spatial slices (which are spacelike in both cases) but about the *depth of the causal future* available for dynamical evolution.

4 The Equivalence Principle and Local Quantum Field Theory

4.1 Local physics is standard

The Einstein equivalence principle [16] guarantees that in a sufficiently small neighborhood of any event—including events inside the horizon—the metric can be approximated by the Minkowski metric, and physics is locally indistinguishable from special relativity.

We **affirm** this principle. Concretely: a freely falling observer inside the horizon of a large black hole can set up Fermi normal coordinates, conduct quantum experiments, maintain entangled states, and observe standard gauge boson propagation—provided the spatial and temporal extent of the experiment is small compared to the curvature scale and, crucially, small compared to the remaining proper time τ_{coh}.

For a supermassive black hole ($M \sim 10^9\, M_\odot$), τ_{coh} at the horizon is approximately 5 hours. Local quantum experiments with duration $\ll 5$ hours proceed normally. There is no instantaneous violation of quantum mechanics at the horizon crossing.

4.2 The global constraint: finite lifetime of all interior processes

What the interior lacks is not local physics but *permanence*. In the exterior, a bound state (an atom, a proton, a star) can exist indefinitely: the infinite causal future guarantees unlimited time for the periodic gauge boson exchanges that maintain its structure. In the interior, every process has a finite remaining lifetime bounded by $\tau_{\text{coh}}(\mathcal{R})$.

This introduces a *viability condition* for extended objects:

Definition 4.1 (Causal viability)**.** *A physical system with characteristic dynamical timescale T (the minimum time required for one complete cycle of its internal dynamics—e.g., one orbital period of an electron, one exchange of a binding gauge boson) is* ***causally viable*** *at areal radius $\mathcal{R}$ if*

$$\tau_{\text{coh}}(\mathcal{R}) \geq N_{\min} T\,, \tag{8}$$

where $N_{\min}$ is the minimum number of dynamical cycles required to sustain the system (at minimum, $N_{\min} = 1$).

In the exterior, $\tau_{\text{coh}} = \infty$, so condition (8) is trivially satisfied for all systems. In the interior, the condition becomes progressively restrictive:

- At $\mathcal{R}$ slightly below r_{s} (large τ_{coh}), only systems with very long dynamical timescales (galaxies, large-scale structures) fail the viability condition.
- As $\mathcal{R} \to 0$ ($\tau_{\text{coh}} \to 0$), every system fails, no matter how fast its dynamics.

This is a hierarchy of *available causal diamonds*, not a failure of local QFT.

4.3 Objects do not dissolve instantaneously; they become unsustainable

We replace the language of "dissolution" used in earlier drafts with the more precise statement:

> *An object crossing the horizon does not instantaneously lose coherence. Rather, it enters a regime where its remaining lifetime is finite and shrinking. When the remaining time drops below the object's dynamical timescale, it can no longer complete even one cycle of its internal dynamics and ceases to exist as a bound system. At the singularity, every system—no matter how fast—becomes unsustainable.*

This respects the equivalence principle (local physics is fine) while identifying the genuine, coordinate-invariant constraint that the interior imposes.

5 The Hierarchy: Gravity and Quantum Mechanics

5.1 What survives at the singularity

We now ask: as $\mathcal{R} \to 0$ and $\tau_{\text{coh}} \to 0$, what physical structures survive?

Survives at all $\mathcal{R} > 0$:

- The spacetime manifold, the metric $g_{\mu\nu}$, and the Riemann curvature $R^{\alpha}{}_{\beta\gamma\delta}$ (all finite for $\mathcal{R} > 0$).
- Geodesics (both timelike and null).
- The Einstein field equations ($R_{\mu\nu} = 0$ in vacuum).

These are all quantities of general relativity. They require only the metric at each point and its local derivatives. They do not require that physical processes have unlimited time to operate.

Requires $\tau_{\text{coh}} > T$ for dynamical timescale T:

- Bound quantum states (atoms, nuclei, hadrons) whose existence depends on periodic gauge boson exchange.
- Coherent quantum fields over regions of size L, which require $\tau_{\text{coh}} > L/c$.

- Entangled systems whose correlations must be verified or maintained through future interactions.

These all involve quantum mechanics and gauge interactions, and all require sufficient causal future to sustain at least one cycle of their dynamics.

5.2 The hierarchy

This leads to the hierarchy:

$$\boxed{\text{Gravity (geometry)} \longrightarrow \text{Causal structure} \longrightarrow \text{Quantum mechanics} \longrightarrow \text{Forces, matter.}} \tag{9}$$

General relativity provides the metric at each point; the metric determines the causal structure (light cones, causal diamonds); the causal structure determines whether quantum mechanics has sufficient arena to operate (causal diamonds large enough for coherent dynamics); quantum mechanics, where it operates, gives rise to gauge forces and matter.

This hierarchy does not refute either theory. General relativity is valid everywhere (including the interior). Quantum mechanics is valid locally everywhere (by the equivalence principle). The hierarchy concerns their *global domains*: only in regions with infinite (or sufficiently large) causal diamonds does quantum mechanics produce stable, long-lived, complex structures.

5.3 On the quantizability of gravity

The hierarchy suggests—but does not prove—that gravity may not be quantizable. The argument:

(i) Quantum mechanics requires a causal structure to define coherent states and evolution.

(ii) The causal structure is determined by the metric, the dynamical variable of gravity.

(iii) Quantizing the metric means placing the causal structure in superposition.

(iv) A superposition of causal structures is a superposition of the conditions under which superposition is defined.

We present this as evidence for a conjecture, not as a theorem. The Schwarzschild interior provides an existence proof of a regime where gravity operates and quantum mechanics has only a restricted (local, finite-duration) role, which is consistent with this conjecture but does not establish it uniquely.

6 Nothingness as the Limit of Vanishing Causal Diamonds

Definition 6.1 (Nothingness)**.** *A spacetime event p is in a state of* ***nothingness*** *if the maximal causal diamond in the future of p has zero four-volume: no physical process of any duration can be completed in the causal future of p.*

By this definition:

- The singularity $\mathcal{R} = 0$ is *absolute nothingness*: $\tau_{\text{coh}} = 0$.
- The interior at $\mathcal{R}_0 < r_{\text{s}}$ is *approaching nothingness*: the maximal causal diamond has finite and shrinking volume.
- The exterior $\mathcal{R} > r_{\text{s}}$ is *not nothingness*: causal diamonds of arbitrary size exist.
- The horizon $\mathcal{R} = r_{\text{s}}$ is the *onset*: the boundary beyond which every process has a finite remaining lifetime.

Nothingness is not the absence of spacetime. The manifold, metric, and curvature persist. It is the absence of the causal depth required for any physical process to occur. It is distinct from the quantum vacuum [10], which is a state *within* quantum mechanics. Nothingness is the condition under which even the quantum vacuum has insufficient future to be defined.

7 The Covariant Causal Gauge Theory

We now construct a dynamical framework for the transition from nothingness (interior) to being (exterior). The key requirement, identified by earlier criticism, is *manifest general covariance*: all couplings must be through geometric scalars, not coordinate-dependent metric components.

7.1 The trapped/untrapped scalar

Definition 7.1 (The causal indicator scalar)**.** *In a spherically symmetric spacetime, let $\mathcal{R}$ denote the areal radius, defined geometrically by $\mathcal{R}^2 = A/(4\pi)$ where A is the area of the orbit two-sphere. Define*

$$\mathcal{X} \equiv g^{\mu\nu} \nabla_\mu \mathcal{R} \, \nabla_\nu \mathcal{R} \, . \tag{10}$$

This is a **coordinate scalar**: it transforms as a scalar under general coordinate transformations, because $\mathcal{R}$ is a geometrically defined scalar field and $\nabla_\mu \mathcal{R}$ is a covector.

In the Schwarzschild geometry, in any coordinate system:

$$\mathcal{X} = f(r) = 1 - \frac{r_{\text{s}}}{r} \, . \tag{11}$$

Verification. In Schwarzschild coordinates, $\mathcal{R} = r$, so $\nabla_\mu \mathcal{R} = \delta^r_\mu$, and $\mathcal{X} = g^{rr} = f(r)$.

In Kruskal–Szekeres coordinates $(T, X_K, \theta, \varphi)$, the metric is $ds^2 = (32G^3M^3/(\mathcal{R}c^4))\, e^{-\mathcal{R}/r_{\text{s}}}(-c^2dT^2 + dX_K^2) + \mathcal{R}^2\, d\Omega^2$, with $\mathcal{R}$ defined implicitly by $(c^2T^2 - X_K^2) = -(1 - \mathcal{R}/r_{\text{s}})e^{\mathcal{R}/r_{\text{s}}} \cdot r_{\text{s}}^2$. The computation gives $g^{\mu\nu}\nabla_\mu \mathcal{R}\, \nabla_\nu \mathcal{R} = 1 - r_{\text{s}}/\mathcal{R} = f(\mathcal{R})$, confirming coordinate invariance.

Physical content:

- $\mathcal{X} > 0$ (exterior): $\nabla_\mu \mathcal{R}$ is spacelike. Outgoing light rays can increase $\mathcal{R}$. The region is *untrapped*.
- $\mathcal{X} = 0$ (horizon): $\nabla_\mu \mathcal{R}$ is null. Outgoing light rays are marginally trapped.
- $\mathcal{X} < 0$ (interior): $\nabla_\mu \mathcal{R}$ is timelike. All causal curves have decreasing $\mathcal{R}$. The region is *trapped*.

The scalar $\mathcal{X}$ is the coordinate-invariant diagnostic of trapping.

7.2 Field content

We introduce:

(i) A complex scalar field φ (the **causal scalar**), carrying charge under a local $U(1)_{\mathcal{C}}$ symmetry.

(ii) A gauge field $\mathcal{A}_\mu$ (the **causal mediator**).

This is distinct from the Standard Model Higgs, which breaks $SU(2)_L \times U(1)_Y$. Whether the two are related is deferred to Paper III.

Gauge transformations, covariant derivative, and field strength are standard:

$$D_\mu \varphi = \partial_\mu \varphi - i g_{\mathcal{C}} \mathcal{A}_\mu \varphi \,, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu \,. \tag{12}$$

7.3 The geometry-coupled Lagrangian

The Lagrangian density is

$$\mathscr{L}_{\mathcal{C}} = \sqrt{-g} \left[-\frac{1}{4} \mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu} + (D_\mu \varphi)^* (D^\mu \varphi) - V(|\varphi|; \mathcal{X}) \right] , \tag{13}$$

where the potential depends *covariantly* on the causal indicator scalar $\mathcal{X}$:

$$\boxed{V(|\varphi|; \mathcal{X}) = -\mu^2 \, \mathcal{X} \, |\varphi|^2 + \lambda \, |\varphi|^4 \,, \qquad \mu^2 > 0, \;\; \lambda > 0 \,.} \tag{14}$$

This is the key equation. The coupling $-\mu^2 \mathcal{X} |\varphi|^2$ replaces the ad-hoc $\Theta(r - r_{\rm s}) \sqrt{f(r)}$ ansatz of earlier work. It is:

- *Manifestly covariant*: $\mathcal{X}$ is a geometric scalar.
- *Dynamically motivated*: the sign of the effective mass-squared depends on the trapped/untrapped character of the region.
- *Minimal*: it is the simplest renormalizable coupling of a scalar field to $\mathcal{X}$.

8 The Causal Higgs Mechanism: Stepwise Derivation

8.1 Step 1: The effective mass and the vacuum structure

The potential (14) can be written as

$$V = \left(-\mu^2 \mathcal{X} \right) |\varphi|^2 + \lambda |\varphi|^4 \equiv m_{\rm eff}^2 \, |\varphi|^2 + \lambda |\varphi|^4 \,, \tag{15}$$

where the effective mass-squared is

$$m_{\rm eff}^2(\mathcal{X}) = -\mu^2 \, \mathcal{X} \,. \tag{16}$$

Exterior ($\mathcal{X} > 0$): $m_{\rm eff}^2 = -\mu^2 \mathcal{X} < 0$. The potential has a Mexican hat shape. The origin $\varphi = 0$ is a local maximum. Spontaneous symmetry breaking occurs.

Interior ($\mathcal{X} < 0$): $m_{\rm eff}^2 = -\mu^2 \mathcal{X} > 0$. The potential is a simple bowl. The origin $\varphi = 0$ is the unique minimum. The symmetry is unbroken.

Horizon ($\mathcal{X} = 0$): $m_{\rm eff}^2 = 0$. The potential is $V = \lambda |\varphi|^4$, which has a flat direction at the origin. This is the critical point of the phase transition.

8.2 Step 2: The vacuum expectation value (derived)

In the broken phase ($\mathcal{X} > 0$), minimizing V with respect to $\rho = |\varphi|^2$:

$$\frac{dV}{d\rho} = -\mu^2\mathcal{X} + 2\lambda\rho = 0 \quad \Longrightarrow \quad \rho = \frac{\mu^2\mathcal{X}}{2\lambda}\,. \tag{17}$$

Therefore the vacuum expectation value is

$$\boxed{\langle|\varphi(\mathcal{R})|\rangle = \begin{cases} v\sqrt{\mathcal{X}} = v\sqrt{f(\mathcal{R})}\,, & \mathcal{X} > 0 \;\; (\mathcal{R} > r_{\rm s}) \\ 0\,, & \mathcal{X} \le 0 \;\; (\mathcal{R} \le r_{\rm s}) \end{cases}} \tag{18}$$

where $v = \mu/\sqrt{2\lambda}$.

This is *derived* from minimizing the potential, not postulated. The profile $\langle|\varphi|\rangle = v\sqrt{f(\mathcal{R})}$ in the exterior and $\langle|\varphi|\rangle = 0$ in the interior emerges as the minimum-energy configuration of the covariantly coupled potential.

Properties:

(a) $\mathcal{R} \to \infty$: $f \to 1$, so $\langle|\varphi|\rangle \to v$ (full condensate).

(b) $\mathcal{R} \to r_{\rm s}^+$: $f \to 0^+$, so $\langle|\varphi|\rangle \to 0$ (condensate vanishes at the horizon).

(c) $\mathcal{R} \le r_{\rm s}$: $\mathcal{X} \le 0$, so $\langle|\varphi|\rangle = 0$ (interior is in the symmetric phase).

Remark 8.1. *The transition is continuous: $\langle|\varphi|\rangle$ is continuous at $\mathcal{R} = r_{\rm s}$ (both sides give zero), though $d\langle|\varphi|\rangle/d\mathcal{R}$ diverges there. This is characteristic of a second-order phase transition. For a large black hole, the VEV changes slowly: $|d\langle|\varphi|\rangle/d\mathcal{R}| \sim v/(2r_{\rm s})$ at distances $\gg r_{\rm s}$ from the horizon, consistent with the equivalence principle (no abrupt local change for a freely falling observer).*

8.3 Step 3: Consistency with the equation of motion

We verify that the derived VEV satisfies the Euler–Lagrange equation. For a static, spherically symmetric ansatz $\varphi = \varphi(\mathcal{R})$, the equation of motion from (13) is

$$\Box\varphi + \mu^2\mathcal{X}\,\varphi - 2\lambda|\varphi|^2\varphi = 0\,, \tag{19}$$

plus terms from the variation of $\mathcal{X}$ with respect to φ, which vanish since $\mathcal{X}$ depends only on the background metric, not on φ.

Substituting $\varphi = v\sqrt{f(\mathcal{R})}$, the potential terms give

$$\mu^2\mathcal{X}\,\varphi - 2\lambda\varphi^3 = \mu^2 f \cdot v\sqrt{f} - 2\lambda\, v^3\, f^{3/2} = v\, f^{3/2}(\mu^2 - 2\lambda v^2) = 0\,, \tag{20}$$

since $v^2 = \mu^2/(2\lambda)$.

The kinetic term $\Box\varphi$ does *not* vanish for $\varphi = v\sqrt{f}$. In the Schwarzschild geometry:

$$\Box(v\sqrt{f}) = \frac{v}{\sqrt{-g}}\partial_\mu\Big(\sqrt{-g}\, g^{\mu\nu}\partial_\nu\sqrt{f}\Big) \neq 0\,. \tag{21}$$

This means the simple potential (14) does not make $v\sqrt{f}$ an *exact* solution. The profile $v\sqrt{f}$ minimizes the potential energy V at each point but does not minimize the total action including gradient energy.

We acknowledge this openly. The derived VEV is the *local potential minimum*, not the exact solution to the full equation of motion. The exact static solution would differ from $v\sqrt{f}$ by corrections that account for the gradient energy. Computing this correction requires solving the nonlinear ODE (19) numerically or adding higher-order curvature couplings to the potential. We leave this to future work and proceed with $v\sqrt{f}$ as the leading-order approximation, valid far from the horizon where $\Box\sqrt{f} \to 0$.

8.4 Step 4: The gauge boson mass

Expanding around the local VEV in unitary gauge ($\varphi = \langle|\varphi|\rangle + h$, Goldstone absorbed), the covariant derivative generates a mass term:

$$|D_\mu \varphi|^2 \;\supset\; g_{\mathcal{C}}^2 \, \langle|\varphi|\rangle^2 \, \mathcal{A}_\mu \mathcal{A}^\mu \,. \tag{22}$$

Comparing with the Proca mass term, the position-dependent gauge boson mass is

$$\boxed{m_{\mathcal{A}}^2(\mathcal{R}) = \frac{2\hbar^2 g_{\mathcal{C}}^2 v^2}{c^2} \, f(\mathcal{R}) \,, \qquad \mathcal{R} > r_{\rm s} \,.} \tag{23}$$

Far from BH ($\mathcal{R} \gg r_{\rm s}$): $m_{\mathcal{A}} \to m_\infty = \hbar g_{\mathcal{C}} v \sqrt{2}/c$, a constant.
At horizon ($\mathcal{R} = r_{\rm s}$): $m_{\mathcal{A}} = 0$. Symmetry restored.
Interior ($\mathcal{R} < r_{\rm s}$): $\langle|\varphi|\rangle = 0$, so $m_{\mathcal{A}} = 0$.

8.5 Step 5: The Higgs (radial) mode mass

Expanding V around the local VEV to second order in h:

$$\begin{aligned} V(\langle|\varphi|\rangle + h) &= -\mu^2 f(\langle|\varphi|\rangle + h)^2 + \lambda(\langle|\varphi|\rangle + h)^4 \\ &= V(\langle|\varphi|\rangle) + \underbrace{\left(-2\mu^2 f \, \langle|\varphi|\rangle + 4\lambda \langle|\varphi|\rangle^3\right)}_{=0 \text{ at minimum}} h + \left(-\mu^2 f + 6\lambda \langle|\varphi|\rangle^2\right) h^2 + \cdots \end{aligned} \tag{24}$$

Using $\langle|\varphi|\rangle^2 = v^2 f = \mu^2 f/(2\lambda)$:

$$m_h^2(\mathcal{R}) = \frac{2}{c^2}\left(-\mu^2 f + 6\lambda \cdot \frac{\mu^2 f}{2\lambda}\right) = \frac{2\mu^2 f}{c^2}\,(3-1) = \frac{4\mu^2 \, f(\mathcal{R})}{c^2} \,, \qquad \mathcal{R} > r_{\rm s} \,. \tag{25}$$

The Higgs mode mass is positive throughout the exterior (where $f > 0$) and vanishes at the horizon ($f = 0$), confirming the phase transition.

8.6 Step 6: Summary of the mechanism

(i) **Exterior ($\mathcal{X} > 0$):** The $U(1)_{\mathcal{C}}$ symmetry is spontaneously broken. The gauge boson acquires mass $m_{\mathcal{A}} \propto \sqrt{f}$. The Higgs mode has mass $m_h \propto \sqrt{f}$. These masses reach their asymptotic values far from the black hole.

(ii) **Horizon ($\mathcal{X} = 0$):** The critical point. All masses vanish. The phase transition occurs continuously.

(iii) **Interior ($\mathcal{X} < 0$):** The symmetric phase. $\langle|\varphi|\rangle = 0$. No causal condensate. Only gravity operates at the fundamental level (though local QFT remains valid transiently, consistent with the equivalence principle and the finite remaining causal future).

9 Mass Creation at the Horizon: Phenomenological Framework

9.1 The gradient energy of the condensate

The VEV (18) varies with $\mathcal{R}$, and this variation stores gradient energy. From $\langle|\varphi|\rangle = v\sqrt{f}$ for $\mathcal{R} > r_{\rm s}$:

$$\frac{d\langle|\varphi|\rangle}{d\mathcal{R}} = \frac{v\, r_{\rm s}}{2\mathcal{R}^2\sqrt{f(\mathcal{R})}}\,. \tag{26}$$

The gradient energy density is

$$\rho_{\rm grad}(\mathcal{R}) = \frac{1}{2}\left(\frac{d\langle|\varphi|\rangle}{d\mathcal{R}}\right)^2 = \frac{v^2\, r_{\rm s}^2}{8\,\mathcal{R}^4\, f(\mathcal{R})}\,, \tag{27}$$

which diverges as $\mathcal{R} \to r_{\rm s}^+$, indicating concentration of energy at the phase boundary.

Integrating with the proper volume element $dV = 4\pi\mathcal{R}^2\, d\mathcal{R}/\sqrt{f}$ and a Planck-scale cutoff near the horizon:

$$E_{\rm grad} \sim \frac{\pi v^2}{r_{\rm s}}\sqrt{\frac{r_{\rm s}}{\ell_{\rm P}}}\,. \tag{28}$$

This energy is concentrated in a thin shell of proper thickness $\sim\sqrt{\ell_{\rm P} r_{\rm s}}$ near the horizon.

9.2 The static solution paradox and the need for dynamics

The preceding calculation was performed on the *static, eternal* Schwarzschild background. On this background, the VEV profile is time-independent, the stress-energy tensor has no energy flux component ($T^{0r} = 0$), and the ADM mass is constant. *There is no mechanism for mass creation within a strictly static framework.*

We acknowledge this explicitly: a mass creation rate **cannot be derived** from the static Schwarzschild solution. To obtain $\dot{M} \neq 0$, one needs a dynamical spacetime—for example, a Vaidya-type metric [20] with $M = M(v)$ along advanced time v, or a collapsing shell geometry—coupled to the scalar field equations.

9.3 The phenomenological ansatz

Nevertheless, we can *parameterize* the expected growth rate using the only available geometric scale. For a quasi-static process (slowly growing horizon), the surface gravity

$$\kappa = \frac{c^4}{4GM} = \frac{c^2}{2r_{\rm s}} \tag{29}$$

sets the rate of change of the redshift factor at the horizon, which in our framework is the rate of change of the causal condensate. On dimensional grounds:

Postulate 1 (Mass creation ansatz)**.** *For a slowly evolving black hole described by $M(t)$, the rate of mass creation by the causal Higgs mechanism is*

$$\dot{M} = \beta\,\kappa^2 = \frac{\beta\, c^8}{16\, G^2\, M^2}\,, \tag{30}$$

where β is an undetermined constant encoding the efficiency of the mechanism, with dimensions [mass^3 length^{-2} time].

This is a *phenomenological hypothesis*, not a derivation from the static background. Its validation or falsification requires solving the coupled Einstein–scalar system on a dynamical spacetime, which we defer to future work. We explore its consequences below.

9.4 Consequences of the ansatz

Horizon growth law. From $r_\mathrm{s} = 2GM/c^2$, integrating $\dot{r}_\mathrm{s} = (2G/c^2)\dot{M}$:

$$r_\mathrm{s}^2\, dr_\mathrm{s} = \frac{\beta c^2 G}{2}\, dt\,, \qquad \Longrightarrow \qquad r_\mathrm{s}^3(t) = r_\mathrm{s}^3(0) + \frac{3\beta c^2 G}{2}\, t\,. \tag{31}$$

At late times, $r_\mathrm{s}(t) \propto t^{1/3}$: self-regulating growth.

Inverse scaling. $\dot{M} \propto M^{-2}$: smaller black holes grow faster, larger ones slower.

10 On Hawking Radiation in This Framework

10.1 The standard derivation

Hawking's result [12] derives particle creation at the horizon by defining a quantum field on the full spacetime, assuming a regular Hadamard vacuum state at and near the horizon [17], and computing the Bogoliubov transformation between in- and out-modes.

10.2 Framework-dependent assumptions

In our framework, the horizon is a causal phase boundary where the effective potential (14) undergoes a second-order phase transition. The causal condensate, gauge boson mass, and Higgs mode mass all vanish at $\mathcal{R} = r_\mathrm{s}$.

The Hawking derivation traces quantum field modes from the exterior through the horizon into the interior. In our framework, the scalar field is in the symmetric phase ($\langle|\varphi|\rangle = 0$) in the interior. Whether the standard Bogoliubov calculation remains valid through a phase transition in the causal sector is an open question.

We do not claim to have *disproved* Hawking radiation. Rather, we observe that within our framework, the assumption of a regular quantum vacuum at the phase boundary requires justification that has not yet been provided. This is a point where our framework makes a *different prediction* from the standard one: in our picture, the isolated Schwarzschild black hole grows (by consuming the condensate), while in the standard picture, it shrinks (by Hawking emission). This difference is in principle observationally testable.

11 Information

11.1 Information and the causal condensate

In our framework, the physical information content of a region is tied to the presence of the causal condensate. We define:

Definition 11.1 (Causal information capacity)**.** *The causal information capacity of a spacelike region Σ in the exterior is*

$$\mathcal{I}[\Sigma] = \frac{1}{4\ell_\mathrm{P}^2}\int_\Sigma \frac{\langle|\varphi|\rangle^2}{v^2}\, dA = \frac{1}{4\ell_\mathrm{P}^2}\int_\Sigma f(\mathcal{R})\, dA\,. \tag{32}$$

At $\mathcal{R} \to \infty$: $\mathcal{I}$ reduces to the area in Planck units. At $\mathcal{R} = r_s$: $\mathcal{I} = 0$. In the interior: $\mathcal{I} = 0$.

The Bekenstein–Hawking entropy $S_{\rm BH} = k_{\rm B} A/(4\ell_{\rm P}^2)$ [15] acquires an interpretation: it counts the Planck-area cells at the horizon where the causal condensate has been reduced to zero.

11.2 The information paradox

If information is encoded in quantum correlations, and quantum correlations require the causal condensate, then matter crossing the horizon passes from a region where information can be sustained ($\langle|\varphi|\rangle > 0$) to one where it cannot ($\langle|\varphi|\rangle = 0$ at the horizon, with only transient local QFT validity in the interior).

The standard information paradox [13] assumes unitarity must hold globally, including across the horizon. In our framework, unitarity is a property of quantum mechanics operating on a causal background. At the horizon—the boundary of the causal condensate—the global applicability of unitarity is in question. We suggest that information is not conserved across the phase boundary, but this is a consequence of the restricted domain of quantum mechanics, not a violation of quantum mechanics within its domain.

12 The Causality Destruction Theorem

12.1 Setup

Define the total causal content of the exterior as

$$\mathcal{C}(M) = 4\pi v^2 \int_{r_s}^{R} \mathcal{R}^2 \sqrt{f(\mathcal{R})}\, d\mathcal{R}\,, \tag{33}$$

where R is an infrared cutoff and $f = 1 - r_s/\mathcal{R}$.

12.2 The theorem

Theorem 12.1 (Causality destruction)**.** *If the black hole mass increases ($\dot{M} > 0$), then $d\mathcal{C}/dt < 0$.*

Proof. We differentiate $\mathcal{C}$ with respect to r_s using Leibniz's rule. The boundary term at r_s vanishes because $f(r_s) = 0$. The integrand contribution is

$$\frac{\partial}{\partial r_s}\sqrt{1 - \frac{r_s}{\mathcal{R}}} = \frac{-1}{2\mathcal{R}\sqrt{1 - r_s/\mathcal{R}}} < 0\,, \qquad \mathcal{R} > r_s\,. \tag{34}$$

Therefore $d\mathcal{C}/dr_s < 0$. Since $\dot{r}_s = (2G/c^2)\dot{M} > 0$,

$$\frac{d\mathcal{C}}{dt} = \frac{d\mathcal{C}}{dr_s} \cdot \dot{r}_s < 0\,. \tag{35}$$

□

Physical content. As the horizon grows, the causal condensate $\langle|\varphi|\rangle = v\sqrt{f(\mathcal{R})}$ weakens at every point in the exterior. The factor $\sqrt{1 - r_s/\mathcal{R}}$ decreases at every fixed $\mathcal{R}$ when r_s increases. The black hole does not merely consume spacetime at its boundary; it dilutes the causal condensate throughout the entire exterior.

Combined with the area theorem ($\delta A \geq 0$, [14]):

> *The total nothingness (measured by horizon area) never decreases. Every increase in horizon area represents an irreversible conversion of causally structured spacetime into the domain of nothingness.*

13 Summary

Result	**Expression**	**Status**
Finite causal diamonds in interior	$\tau_{\text{coh}} \to 0$ as $\mathcal{R} \to 0$	Theorem (GR)
Local QFT valid inside horizon	Equivalence principle	Standard physics
Global constraint: finite lifetime	$\tau_{\text{coh}} < \infty$ in interior	Theorem (GR)
Covariant causal scalar $\mathcal{X}$	$g^{\mu\nu}\nabla_\mu\mathcal{R}\,\nabla_\nu\mathcal{R}$	Geometric def.
VEV from potential minimum	$v\sqrt{f(\mathcal{R})}$ for $\mathcal{R} > r_s$, 0 otherwise	Derived
Gauge boson mass	$\propto v\sqrt{f}$, vanishes at r_s	Derived
Mass creation rate	$\beta c^8/(16G^2M^2)$	Phenomenological
Causality destruction	$d\mathcal{C}/dt < 0$ when $\dot{M} > 0$	Theorem

14 Discussion

14.1 What this framework does and does not claim

We do not modify general relativity. The Schwarzschild solution is the starting point.

We do not reject quantum mechanics. The equivalence principle guarantees local QFT validity everywhere, including inside the horizon. What we identify is a global constraint: the finite causal future of the interior limits the sustainability of quantum structures.

We do not claim to have disproved Hawking radiation. We have identified a framework-dependent assumption in the standard derivation (the existence of a regular vacuum at the causal phase boundary) and noted that our framework makes a different prediction. The resolution is empirical.

What we propose is a hierarchy—gravity → causal structure → quantum mechanics—supported by the observation that general relativity operates in regimes (the Schwarzschild interior, especially near the singularity) where quantum mechanics has progressively restricted scope.

14.2 What the covariant formulation achieves

The use of $\mathcal{X} = g^{\mu\nu}\nabla_\mu\mathcal{R}\,\nabla_\nu\mathcal{R}$ resolves the covariance objection to earlier formulations. The VEV profile is *derived* from the potential minimum, not postulated. The phase transition at $\mathcal{X} = 0$ is geometric: it coincides with the trapped/untrapped transition, which is the coordinate-invariant definition of the horizon.

The remaining theoretical gap is the equation-of-motion correction: the profile $v\sqrt{f}$ is the potential minimum at each point but not the exact static solution of the full field equation (which would include gradient corrections). Computing this correction is a concrete, well-defined problem.

14.3 What the mass creation ansatz needs

The postulated mass creation rate $\dot{M} = \beta\kappa^2$ is phenomenological. It cannot be derived from a static background. A proper derivation requires coupling the scalar field to a dynamical spacetime (e.g., Vaidya [20]) and computing the stress-energy flux across the horizon. This is the most important open calculation in the framework.

14.4 Testable predictions

(i) **No Hawking radiation from isolated black holes.** Observation of Hawking radiation would falsify our framework.

(ii) **No quantum gravitational signatures.** The BMV experiment [24, 25] should yield null results if gravity is not quantizable.

(iii) **Gravitational wave imprints.** During mergers, rapid changes in horizon area may encode information about the causal condensate dynamics in the quasi-normal mode spectrum.

14.5 Open problems

(i) Solving the full static equation of motion for $\varphi(\mathcal{R})$ with the potential (14) on the Schwarzschild background.

(ii) Coupling to a dynamical spacetime (Vaidya) to derive $\dot{M}$ from first principles.

(iii) Determining β from the microscopic parameters $g_{\mathcal{C}}$, μ, λ.

(iv) Extension to Kerr.

(v) The emergence of $SU(3) \times SU(2) \times U(1)$ from the causal phase transition.

(vi) The dark sector as memory effects of causal connectedness.

(vii) The emergence of time from simultaneity.

15 Conclusion

The Schwarzschild black hole reveals a hierarchy. Inside the horizon, every physical process has a finite remaining lifetime that shrinks to zero at the singularity. The equivalence principle guarantees that local quantum field theory remains valid, but the global capacity to sustain stable structures, indefinite coherence, and permanent bound states requires infinite causal future—which exists only in the exterior.

General relativity operates on both sides of the horizon. It requires only the local metric. Quantum mechanics operates on both sides locally, but its global fruits—stable atoms, propagating gauge bosons over unlimited time, information stored indefinitely—require the exterior's infinite causal diamonds.

At the horizon, a covariantly defined scalar $\mathcal{X} = g^{\mu\nu}\nabla_\mu\mathcal{R}\,\nabla_\nu\mathcal{R}$ changes sign. This drives a phase transition in a causal scalar field: the $U(1)_\mathcal{C}$ symmetry is broken in the exterior (where $\mathcal{X} > 0$) and restored in the interior (where $\mathcal{X} < 0$). The vacuum expectation value, gauge boson mass, and Higgs mode mass are all derived from the potential minimum, not postulated.

The horizon is the boundary between the region where physics can build lasting structures and the region where every structure has a countdown to its end. At the singularity, the countdown reaches zero. Nothing can survive.

From nothingness, everything.

Acknowledgments

D.S. acknowledges support from the University of Luxembourg. A.K. acknowledges support from the Université de Toulouse.

References

[1] K. Schwarzschild, *Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie*, Sitzungsber. Preuss. Akad. Wiss. (1916) 189.

[2] M. D. Kruskal, *Maximal Extension of Schwarzschild Metric*, Phys. Rev. **119** (1960) 1743.

[3] R. Penrose, *Gravitational Collapse and Space-Time Singularities*, Phys. Rev. Lett. **14** (1965) 57.

[4] B. S. DeWitt, *Quantum Theory of Gravity. I. The Canonical Theory*, Phys. Rev. **160** (1967) 1113.

[5] G. 't Hooft and M. Veltman, *One-Loop Divergencies in the Theory of Gravitation*, Ann. Inst. H. Poincaré A **20** (1974) 69.

[6] J. Polchinski, *String Theory*, Vols. 1–2, Cambridge University Press, 1998.

[7] C. Rovelli, *Quantum Gravity*, Cambridge University Press, 2004.

[8] M. Reuter, *Nonperturbative Evolution Equation for Quantum Gravity*, Phys. Rev. D **57** (1998) 971.

[9] F. Dyson, *Is a Graviton Detectable?*, Int. J. Mod. Phys. A **28** (2013) 1330041.

[10] L. M. Krauss, *A Universe from Nothing*, Free Press, 2012.

[11] P. W. Higgs, *Broken Symmetries and the Masses of Gauge Bosons*, Phys. Rev. Lett. **13** (1964) 508.

[12] S. W. Hawking, *Particle Creation by Black Holes*, Commun. Math. Phys. **43** (1975) 199.

[13] S. W. Hawking, *Breakdown of Predictability in Gravitational Collapse*, Phys. Rev. D **14** (1976) 2460.

[14] J. M. Bardeen, B. Carter, and S. W. Hawking, *The Four Laws of Black Hole Mechanics*, Commun. Math. Phys. **31** (1973) 161.

[15] J. D. Bekenstein, *Black Holes and Entropy*, Phys. Rev. D **7** (1973) 2333.

[16] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.

[17] R. M. Wald, *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*, University of Chicago Press, 1994.

[18] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press, 1982.

[19] W. G. Unruh, *Notes on Black-Hole Evaporation*, Phys. Rev. D **14** (1976) 870.

[20] P. C. Vaidya, *The Gravitational Field of a Radiating Star*, Proc. Indian Acad. Sci. A **33** (1951) 264.

[21] L. Susskind, L. Thorlacius, and J. Uglum, *The Stretched Horizon and Black Hole Complementarity*, Phys. Rev. D **48** (1993) 3743.

[22] A. Almheiri, D. Marolf, J. Polchinski, and J. Sully, *Black Holes: Complementarity vs. Firewalls*, JHEP **02** (2013) 062.

[23] R. Bousso, *The Holographic Principle*, Rev. Mod. Phys. **74** (2002) 825.

[24] S. Bose *et al.*, *Spin Entanglement Witness for Quantum Gravity*, Phys. Rev. Lett. **119** (2017) 240401.

[25] C. Marletto and V. Vedral, *Gravitationally Induced Entanglement between Two Massive Particles is Sufficient Evidence of Quantum Effects in Gravity*, Phys. Rev. Lett. **119** (2017) 240402.

[26] S. A. Hayward, *General Laws of Black-Hole Dynamics*, Phys. Rev. D **49** (1994) 6467.

[27] H. Kodama, *Conserved Energy Flux for the Spherically Symmetric System*, Prog. Theor. Phys. **63** (1980) 1217.